
P M - 0 4 - P S 0 3

Mzansi Bank FICA Compliance Automation

Robotic Process Automation Solution

ASSESSMENT EVIDENCE PACKAGE

<https://github.com/TheFish1011/PM-04-PS03.git>

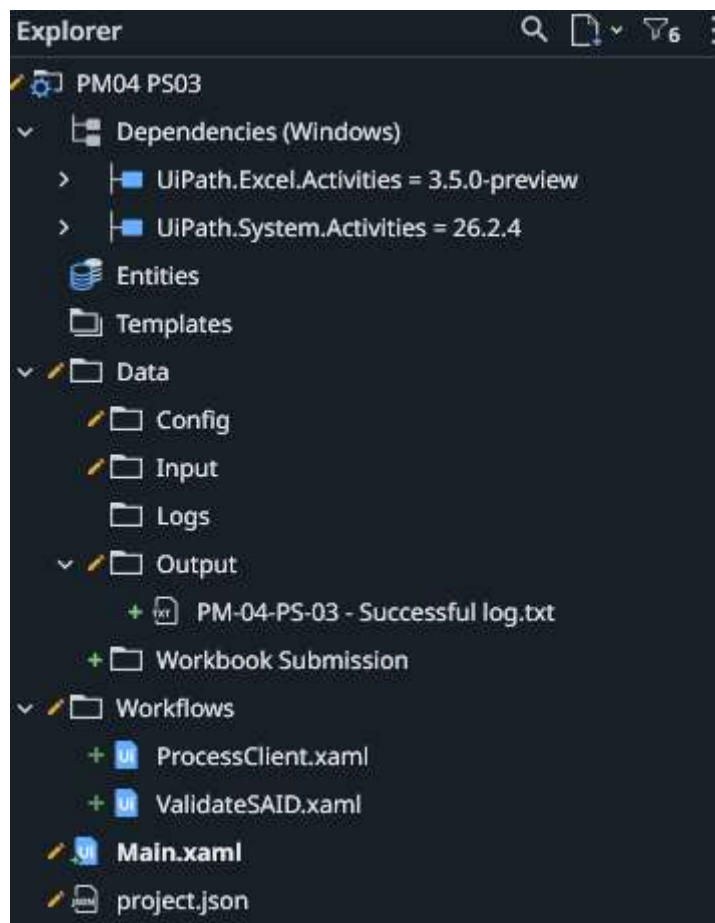
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1. Selected RPA Platform

UiPath was selected due to its strong enterprise adoption within financial institutions, robust security features, and extensive library of prebuilt automation activities. It supports both low-code and custom development, making it suitable for compliance-heavy environments such as banking. Additionally, its integration with Orchestrator enables centralized control, logging, and scheduling.

2. Development Environment Setup

1. Installed UiPath Studio
2. Created new Process project
3. Created workflows: Main.xaml, ProcessClient.xaml, ValidateSAID.xaml
4. Created Config.csv
5. Verified dependencies



3. Initial Workflow

Basic workflow created with config loading and logging.



4. Process Design & Main Workflow

Phase 1 — Configuration Loading

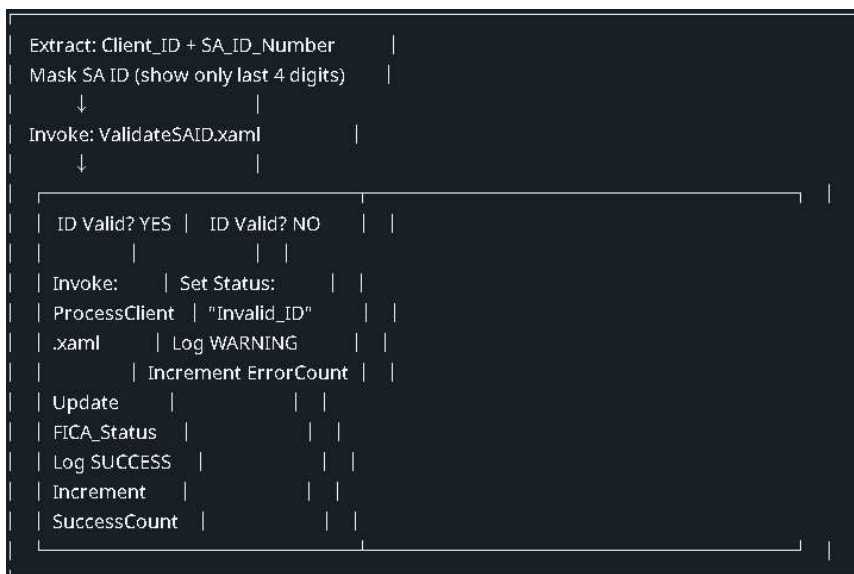
Step	Activity	Description
1.1	Read Config.csv	Reads the configuration file from <code>Config\Config.csv</code>
1.2	Parse Config to DataTable	Parses the raw CSV text into a structured <code>dt_Config</code> DataTable
1.3	Load Config Values	Iterates each row and maps keys to variables:
	→ <code>InputFilePath</code>	Path to the client input CSV
	→ <code>PortalURL</code>	URL of the FICA compliance portal
	→ <code>OutputFolder</code>	Directory where output files will be saved
1.4	Set Log File Path	Constructs the log file path as <code>OutputFolder\processing_log.txt</code>
1.5	Log Config Summary	Logs loaded config values to the Studio output panel

Phase 2 — Data Ingestion (wrapped in Try/Catch)

Step	Activity	Description
2.1	Read FICA CSV as Text	Reads the input CSV from <code>str_InputFilePath</code>
2.2	Parse CSV to DataTable	Converts raw text into <code>dt_ClientData</code> DataTable
2.3	Log Process Start	Logs total record count and timestamps to both the Studio log and the <code>.txt</code> log file

Phase 3 — Client Processing Loop (For Each Client Row)

For every client record in dt_ClientData:



Sub-workflow calls:

- `ValidateSAID.xaml` — Validates the SA ID format/checksum → returns `bool_IsValid`
- `ProcessClient.xaml` — Submits the client to the FICA portal → returns `str_FICAResult`

Phase 4 — Save Results

Step	Activity	Description
4.1	Build CSV Header	Sets header: <code>Client_ID, Full_Name, SA_ID Number, FICA_Status</code>
4.2	Build Output CSV Text	Iterates <code>dt_ClientData</code> and builds the full CSV string
4.3	Write Updated CSV	Writes to <code>OutputFolder\FICA_Compliance_Data_Updated.csv</code>
4.4	Log Completion	Logs final counts (Success & Errors) to both log targets

Exception Handling Strategy

Scope	Exception Type	Handling
Outer	FileNotFoundException	Logs CRITICAL error – process cannot continue, terminates gracefully
Outer	General Exception	Logs unexpected error message
Per Client	General Exception	Sets FICA_Status = "Error", logs ERROR entry, increments ErrorCount – continues to next client
Invalid ID Business Exception		Sets FICA_Status = "Invalid ID", logs WARNING, increments ErrorCount – continues to next client

Data Flow Summary

Config.csv → dt_Config → Runtime Settings

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ClientData.csv → dt_ClientData → Per-Row Processing

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ValidateSAID.xaml —|
ProcessClient.xaml —|

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FICA_Compliance_Data_Updated.csv ←
processing_log.txt ←